

第一类曲面积分

$$\iint_S f(x,y,z) dS = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta S_i$$

曲面 S 由 $z=g(x,y)$ 确定

$$\iint_S f(x,y,z) dS = \iint_D f(x,y,g(x,y)) \sqrt{1+(g'_x)^2+(g'_y)^2} dx dy$$

向量式方程

$$r = (x(u,v), y(u,v), z(u,v))$$

$$\iint_S f(x,y,z) dS = \iint_D f(x(u,v), y(u,v), z(u,v)) |\vec{r}_u \times \vec{r}_v| du dv$$

$$\vec{r}_u \times \vec{r}_v = \left(\frac{\partial(y,z)}{\partial(u,v)}, \frac{\partial(z,x)}{\partial(u,v)}, \frac{\partial(x,y)}{\partial(u,v)} \right)$$

$$\iint_S f(x,y,z) dS = \iint_D f(x(u,v), y(u,v), z(u,v)) \sqrt{EG-F^2} du dv$$

$$E = r'_u \cdot r'_u \quad F = r'_u \cdot r'_v \quad G = r'_v \cdot r'_v$$

$$|\vec{r}_u \times \vec{r}_v| = \sqrt{|\vec{r}_u|^2 \cdot |\vec{r}_v|^2 - (\vec{r}_u \cdot \vec{r}_v)^2}$$

例: 计算 $\iint_S z \, ds$, $S = x^2 + y^2 + z^2 = a^2$ 被 $z = h$ ($0 < h < a$) 所截出的顶部

法一: $\vec{r}(x, y) = (x, y, \sqrt{a^2 - x^2 - y^2}) \quad (x, y) \in D$

$$D = \{(x, y) \mid x^2 + y^2 \leq a^2 - h^2\}$$

$$\vec{r}_x = (1, 0, \frac{-x}{\sqrt{a^2 - x^2 - y^2}}) \quad \vec{r}_y = (0, 1, \frac{-y}{\sqrt{a^2 - x^2 - y^2}})$$

$$|\vec{r}_x \times \vec{r}_y| = \left| \left(\frac{x}{\sqrt{a^2 - x^2 - y^2}}, \frac{y}{\sqrt{a^2 - x^2 - y^2}}, 1 \right) \right|$$
$$= \sqrt{1 + \frac{x^2}{a^2 - x^2 - y^2} + \frac{y^2}{a^2 - x^2 - y^2}} = \frac{a}{\sqrt{a^2 - x^2 - y^2}}$$

$$\text{原式} = \iint_D \sqrt{a^2 - x^2 - y^2} \cdot \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dx \, dy = a\pi(a^2 - h^2)$$

法二: $r = (a \sin \varphi \cos \theta, a \sin \varphi \sin \theta, a \cos \varphi)$

$$D = \{(\varphi, \theta) \mid 0 \leq \varphi \leq \arccos \frac{h}{a}, 0 \leq \theta \leq 2\pi\}$$

$$ds = a^2 \sin \varphi$$

$$\text{原式} = \iint_S z \, ds = 2\pi \int_0^{\arccos \frac{h}{a}} a \cdot \cos \varphi \cdot a^2 \sin \varphi \, d\varphi = \pi a(a^2 - h^2)$$

① 找两个变量 (投影平面)

② 计算 $\vec{r}_x \times \vec{r}_y$

对于 x, y 可用公式

$$\iint_D f(x, y, g(x, y)) \sqrt{1 + (g'_x)^2 + (g'_y)^2} \, dx \, dy$$

例: 空间立体 V 由曲面 $x^2+y^2=1$ 与平面 $z=2-x$ 以及 $z=0$ 所围成, S 为 V 的边界曲面

(1) 求曲面积分 $\iint_S x ds$.

$$S = S_{\text{底}} + S_{\text{侧}} + S_{\text{顶}}$$

$$S_{\text{底}}: \vec{r}(x,y) = (x,y,0) \quad x^2+y^2 \leq 1 \quad \iint_{x^2+y^2 \leq 1} x ds = 0$$

$$S_{\text{侧}}: \vec{r}(\theta, z) = (\cos\theta, \sin\theta, z) \quad 0 \leq \theta \leq 2\pi \quad 0 \leq z \leq 2 - \cos\theta$$

$$\iint x ds = \int_0^{2\pi} d\theta \int_0^{2-\cos\theta} \cos\theta dz = \int_0^{2\pi} \cos\theta (2 - \cos\theta) d\theta = -\pi$$

$$S_{\text{顶}}: \vec{r}(x,y) = (x,y,2-x) \quad r'_x = (1,0,-1) \quad r'_y = (0,1,0)$$

$$\iint_{x^2+y^2 \leq 1} x ds = \iint \sqrt{2} x dx dy = 0$$

$$S = -\pi$$

(2) 若 S 有均匀密度 a , 求 S 的质量 M

$$M = \iint_S a ds = 2 \iint_{S_1} a ds = 2 \iint_{S_2} a ds + 2 \iint_{S_3} a ds + 2 \iint_{S_4} a ds$$

$$= 2(1+\sqrt{2}) \iint_{D_1} a dx dy + 2 \iint_{D_2} \frac{a dx dz}{\sqrt{1-x^2}}$$

$$= (1+\sqrt{2}) a \pi + 2a \int_{-1}^1 \frac{dx}{\sqrt{1-x^2}} \int_0^{2-x} dz$$

$$= (5+\sqrt{2}) a \pi$$

